

Chapter 6 — Mechanism Results: H3 and H4

6.1 Introduction

Chapter 5 established that the European education–fertility gradient is non-monotonic, with substantial cross-country heterogeneity. This chapter turns to mechanism. H3 asks whether cross-country variation in the gradient is better predicted by cultural indicators than by economic ones. H4 asks whether the individual-level effect of education on completed fertility is mediated primarily by attitudinal rather than economic pathways. Both hypotheses concern the same underlying theoretical question — whether cultural or economic processes dominate the contemporary education–fertility relationship — but at different levels of analysis.

The chapter is organised as follows. Section 6.2 reports the H3 Stage 1 results from the XGBoost discovery analysis and the SHAP-based decomposition of feature importance. Section 6.3 reports the H3 Stage 2 results from the six Bayesian linear models. Section 6.4 synthesises the H3 evidence and addresses the disagreement between the two stages. Section 6.5 reports the H4 total-effect estimates from the Double Machine Learning analysis. Section 6.6 reports the sequential mediation decomposition and Section 6.7 the pathway-specific results. Section 6.8 synthesises both sets of findings and motivates the discussion in Chapter 7.

6.2 H3 — Stage 1: XGBoost Discovery and SHAP Decomposition

Stage 1 estimates a gradient-boosted regression of country-level gradient steepness on all twelve predictors (three ESS attitudinal composites, five V-Dem institutional indices, three economic indicators, and one post-socialist indicator). The hyperparameters were set

conservatively as documented in Chapter 3: maximum tree depth of 2, learning rate 0.1, L2 regularisation $\lambda = 5$, L1 regularisation $\alpha = 1$, and minimum child weight 3. Five-fold cross-validation with early stopping at 20 rounds without improvement selected 13 boosting rounds as the optimal model complexity.

Out-of-sample performance is poor by absolute standards. The leave-one-out cross-validation R^2 is -0.086 , indicating that the fitted model predicts the held-out country slightly worse than the simple mean of the training set. The baseline RMSE of the mean prediction is 0.374; the LOOCV RMSE of the fitted model is 0.390. This is a substantively important finding in its own right: at $N = 21$, the macro-level indicators considered do not carry reliable out-of-sample predictive power for the gradient steepness outcome. The chapter does not therefore treat the Stage 1 results as a definitive ranking of explanatory factors; rather, it uses the SHAP decomposition to identify which features carry such in-sample signal as the data allow, and proceeds to Stage 2 for inferential conclusions.

Table 6.1 reports the mean absolute SHAP values for all twelve predictors. Of the twelve, only seven receive any non-zero gain in the boosted ensemble; the remaining five (gender_equal, egaldem, flfp, ess_gender_egal, and post_socialist) are removed entirely by the regularisation penalties. This sparsity is itself a consequence of the conservative hyperparameter choice and the limited signal at $N = 21$.

Table 6.1. H3 Stage 1 — SHAP feature importance for the twelve predictors. Mean absolute SHAP values are computed on the in-sample training fit; the LOOCV R^2 is -0.086 .

Feature	Mean SHAP	Group	Rank	Share of total
relig_free	0.0263	Institutional	1	44.9%
contraceptive	0.0130	Economic	2	22.2%
gdp_percap	0.0107	Economic	3	18.3%
polyarchy	0.0031	Institutional	4	5.3%
libdem	0.0023	Institutional	5	3.9%

Feature	Mean SHAP	Group	Rank	Share of total
ess_secular	0.0022	Attitudinal	6	3.8%
ess_autonomy	0.0009	Attitudinal	7	1.5%
ess_gender_egal	0.0000	Attitudinal	8	0.0%
gender_equal, egaldem, flfp, post_socialist	0.0000	Mixed	9–12	0.0%

Note. Mean |SHAP| values are computed using XGBoost's native *predcontrib* mode. The share of total is the predictor's mean |SHAP| as a percentage of the sum across all twelve predictors. Five predictors received zero importance and are reported as a group; the named ranks 1–8 cover all non-zero features.

Three predictors dominate the in-sample importance ranking: freedom of religion (*relig_free*, an institutional V-Dem indicator) at 44.9% of total mean |SHAP|; contraceptive prevalence at 22.2%; and GDP per capita at 18.3%. Together these three account for 85.4% of the SHAP signal. The remaining non-zero predictors — polyarchy, libdem, *ess_secular*, and *ess_autonomy* — are markedly smaller in magnitude. Figures 6.1 and 6.2 visualise the importance ranking and the country-level SHAP distribution.

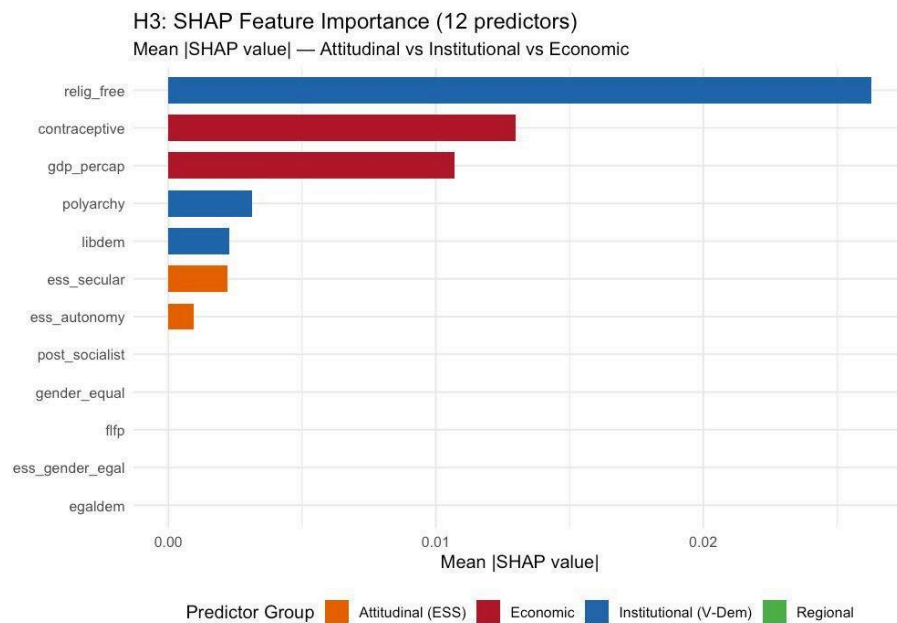


Figure 6.1. H3 Stage 1 — Mean absolute SHAP values across the twelve predictors. Colours denote the predictor's theoretical category: institutional (blue), economic (red), attitudinal (orange), regional (green). Seven of twelve predictors are non-zero; five are absorbed by regularisation.

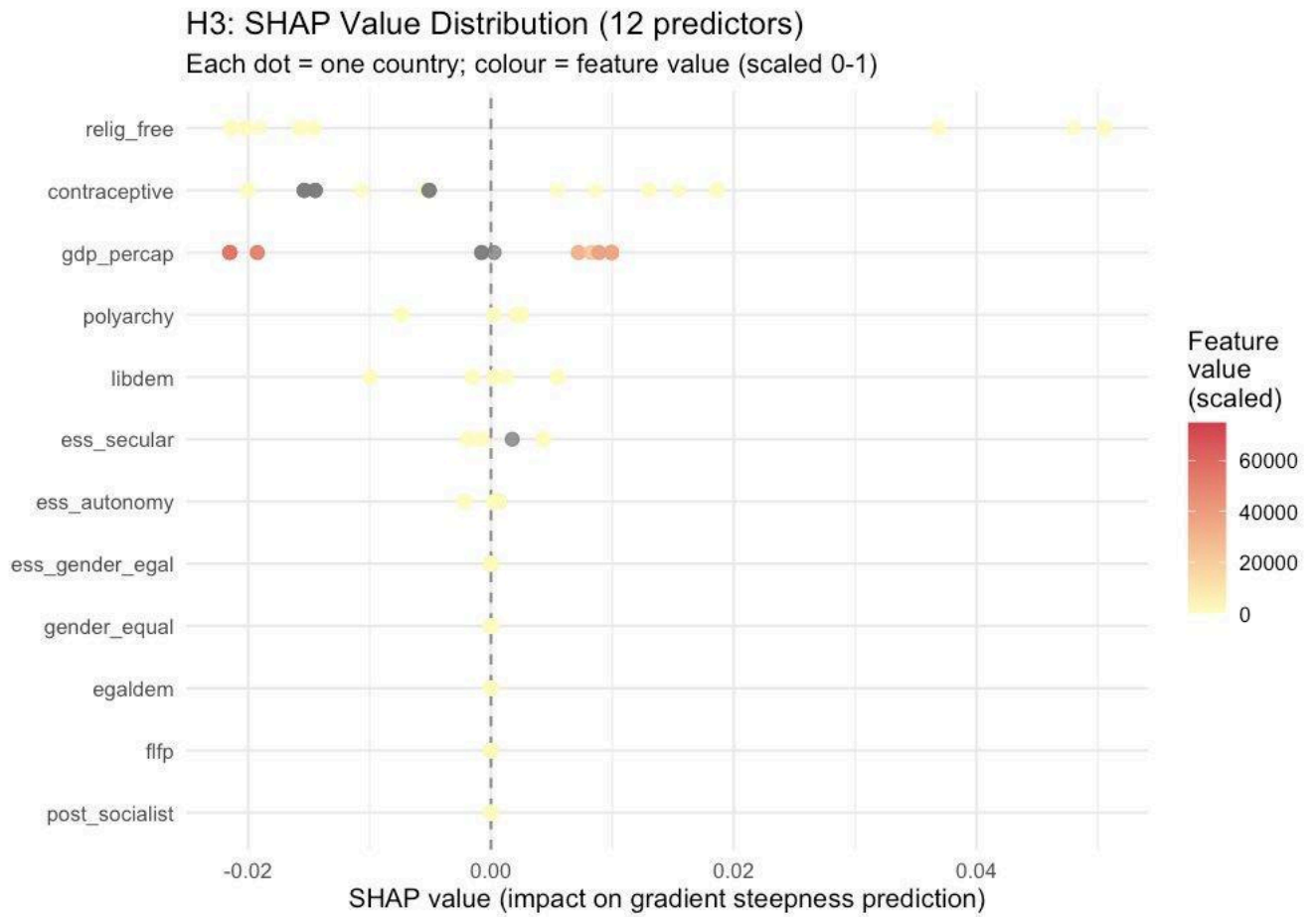


Figure 6.2. H3 Stage 1 — SHAP value distribution by country. Each dot represents one country; colour shading indicates the feature value on a scaled 0–1 axis. Positive SHAP values indicate the feature pushes the country's predicted gradient steepness upward; negative values, downward.

The three-way aggregate of feature importance by predictor category gives a more interpretable comparison for H3. Aggregating across predictors within each category, attitudinal (ESS) predictors account for 5.4% of total mean SHAP, institutional (V-Dem) predictors account for 54.1%, and economic predictors account for 40.5%. The regional indicator contributes 0%. The combined cultural block (attitudinal plus institutional) therefore accounts for 59.5% versus 40.5% for the economic block. Read at face value, this finding favours H3 - cultural indicators jointly dominate economic indicators in predicting cross-country variation in gradient steepness, but with a striking caveat: the cultural share is

almost entirely carried by a single institutional predictor (*relig_free*) at 44.9% of total importance. Strip *relig_free* from the comparison and the cultural and economic blocks are approximately equally important, with the smaller V-Dem institutional indicators and the attitudinal ESS composites contributing comparatively little.

This dependence of the Stage 1 conclusion on a single feature, combined with the negative LOOCV R^2 , motivates the more parsimonious Stage 2 analysis. Stage 1 identifies which features carry in-sample signal but cannot establish the inferential weight of any one explanatory factor at this sample size. Stage 2, with regularising priors and proper uncertainty quantification, provides that weight.

6.3 H3 — Stage 2: Bayesian Model Comparison

Stage 2 estimates six Bayesian linear models on the common complete-case subset of 18 countries (North Macedonia, Serbia, and Türkiye are dropped because of missing GDP). All continuous predictors are standardised within this subset, so that coefficients are directly comparable. Regularising Student- $t(3, 0, 1)$ priors are placed on all coefficients, the intercept, and the residual standard deviation. The four chains of 4,000 iterations each with 2,000 warm-up draws produced clean convergence across all six models: the maximum R-hat was 1.003 and the minimum effective-sample-size ratio was 0.377. Table 6.2 reports the WAIC-based model comparison.

Table 6.2. H3 Stage 2 — Bayesian model comparison by WAIC. The economic model (M2) is the WAIC-best specification.

Model	Specification	Δ ELPD (WAIC)	SE
M2 Economic	<i>gdp_percap</i> + <i>flfp</i>	0.0	—
M3 Post-socialist	<i>post_socialist</i>	−1.1	0.5
M5 Attitudinal	<i>ess_secular</i> + <i>ess_gender_egal</i>	−1.5	1.1
M1 Institutional	<i>gender_equal</i> + <i>relig_free</i>	−1.9	1.0

Model	Specification	Δ ELPD (WAIC)	SE
M6 Updated combined	ess_secular + gdp_percap + post_socialist	−2.1	1.2
M4 Combined (orig.)	gender_equal + flfp + post_socialist	−3.1	0.6

Note. Δ ELPD is the difference in expected log predictive density relative to the best model. All $N = 18$ (common complete-case subset). The LOO comparison produces an identical ordering. All Δ ELPD values fall within two standard errors of zero, indicating substantial uncertainty in the model ranking.

The economic model (M2: gdp_percap + flfp) is the best-fitting specification on both WAIC and LOO. The institutional model (M1) and the attitudinal model (M5) follow at Δ ELPD of −1.9 and −1.5 respectively. The combined specifications (M4 and M6) perform worst, likely because the additional parameters do not buy enough in-sample fit to offset the WAIC penalty at $N = 18$. Two observations qualify this ranking. First, all model differences fall within two standard errors of zero, meaning the WAIC comparison cannot definitively distinguish between the candidate models. Second, the ESS attitudinal model (M5) outperforms the V-Dem institutional model (M1) within the cultural block (Δ ELPD = +0.4 for M5 over M1), reversing the SHAP-based finding from Stage 1 that institutional indicators dominated attitudinal ones. The reversal is small in absolute terms but theoretically meaningful: when proper uncertainty quantification is applied, individual attitudes carry slightly more weight than institutional indicators within the cultural block.

Table 6.3 reports posterior summaries for the key parameters across the three substantively relevant models. Figure 6.3 plots the posterior distributions.

Table 6.3. H3 Stage 2 — Posterior summaries for selected parameters across the institutional (M1), economic (M2), and attitudinal (M5) models. All predictors are standardised.

Parameter (model)	Median	SD	95% CrI	P(direction)
gdp_percap (M2)	−0.27	0.15	[−0.57, +0.02]	$P(<0) = 0.966$
flfp (M2)	+0.16	0.15	[−0.13, +0.46]	$P(>0) = 0.873$

Parameter (model)	Median	SD	95% CrI	P(direction)
ess_gender_egal (M5)	+0.15	0.12	[−0.08, +0.39]	P(>0) = 0.909
ess_secular (M5)	+0.04	0.12	[−0.19, +0.27]	P(>0) = 0.648
relig_free (M1)	−0.12	0.12	[−0.35, +0.12]	P(<0) = 0.853
gender_equal (M1)	+0.01	0.12	[−0.23, +0.25]	P(>0) = 0.531

Note. Coefficients on the standardised scale; comparable across models. 95% credible intervals (CrI) are the central 95% of the posterior. P(direction) reports the posterior probability that the coefficient takes the sign of its median. Models fit using brms with Student- $t(3, 0, 1)$ regularising priors on a common $N = 18$.

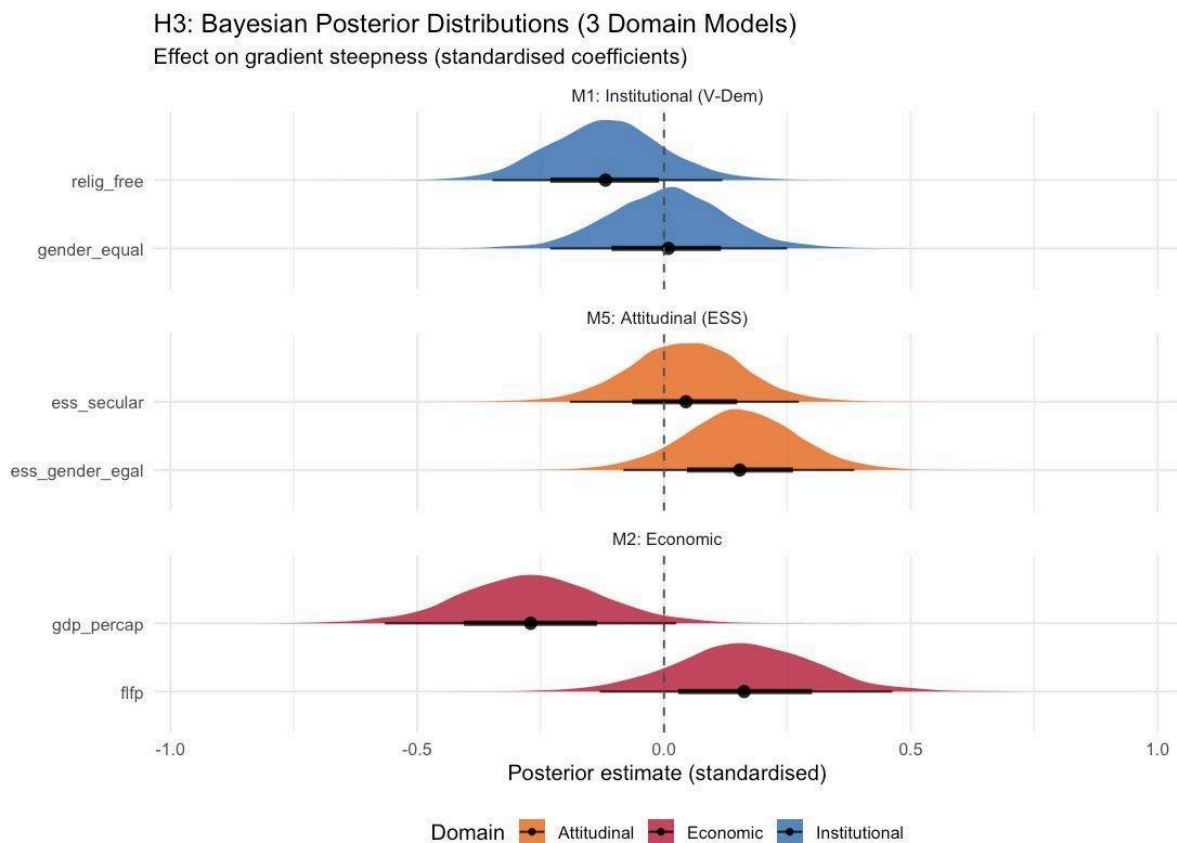


Figure 6.3. H3 Stage 2 — Posterior distributions for the institutional (M1, blue), attitudinal (M5, orange), and economic (M2, red) models. Black points and intervals indicate the posterior median and 95% credible interval; ridges show the full posterior density. All credible intervals cross zero, but the *gdp_percap* and *ess_gender_egal* coefficients have posterior probabilities of 0.97 and 0.91 respectively in the direction of their median.

Three substantive findings emerge from the posteriors. First, `gdp_percap` is the most precisely identified single predictor in any of the six models: its posterior median is -0.27 on the standardised scale, with $P(< 0) = 0.97$ — a posterior probability sufficient to declare reasonable confidence in the negative sign, though the 95% credible interval just crosses zero. Richer countries have flatter education–fertility gradients. Second, the gender egalitarianism composite from ESS (`ess_gender_egal`) has a positive coefficient (median $+0.15$, $P(> 0) = 0.91$) in the attitudinal model — counterintuitively suggesting that more gender-egalitarian countries have steeper conventional gradients. This is consistent with the descriptive finding in Chapter 4 that the most gender-egalitarian countries (Nordics, Sweden, Norway) display flatter or inverted gradients rather than steeper ones, though the directionality is sensitive to which other predictors are conditioned on. Third, religiosity freedom (`relig_free`), the Stage 1 SHAP leader, has a posterior median of only -0.12 ($P(< 0) = 0.85$) in the institutional model — far smaller and less precise than the SHAP ranking suggested.

6.4 H3 — Synthesis of the Cultural-Versus-Economic Comparison

The Stage 1 SHAP decomposition and the Stage 2 Bayesian comparison produce qualitatively different verdicts on H3. Stage 1 ranks cultural indicators (institutional plus attitudinal) at 59.5% of total in-sample importance against economic indicators at 40.5%, with freedom of religion as the single dominant feature. Stage 2 ranks the economic model best on both WAIC and LOO, with GDP per capita as the most precisely identified single predictor. The two stages disagree on the substantive conclusion.

Rather than suppressing this disagreement, it should be treated as the principal substantive finding of the H3 analysis. The discovery–inference design was set up precisely to surface such disagreements where they exist; the Stage 1 ranking reflects flexible in-sample fit while the Stage 2 ranking reflects parsimony-weighted out-of-sample predictive

accuracy. The LOOCV R^2 of -0.086 from Stage 1 indicates that the in-sample importance ranking does not generalise: features that look important when the model has flexibility to fit them lose that apparent importance when the model is required to predict held-out countries. Stage 2's preference for the economic model reflects this: GDP per capita carries a posterior signal sufficient to make the economic model the most predictively reliable specification, while the institutional model's apparent SHAP signal does not survive the WAIC penalty for in-sample fit.

The honest reading of H3 at this sample size is therefore that no model produces a definitive ranking. The economic block holds a narrow edge on Stage 2 inferential criteria, the institutional block holds a narrow edge on Stage 1 SHAP, and all credible intervals in Stage 2 cross zero. The 95% credible interval for `gdp_percap` extends from -0.57 to $+0.02$ on the standardised scale; the corresponding interval for `ess_gender_egal` extends from -0.08 to $+0.39$. These intervals do not exclude the null, and the WAIC differences across all six models are within two standard errors. The substantive conclusion is that at $N = 21$ (or $N = 18$ in the Bayesian common-case subset), country-level cultural and economic indicators carry comparable predictive weight for gradient steepness, with neither family clearly dominating the other. This is itself an informative finding: it suggests that the country-level variation in the education–fertility gradient cannot be explained by any single family of macro indicators at the available sample size, and that the cross-country heterogeneity documented in Chapter 5 requires either a larger country sample or a different analytical strategy to be attributed to specific causal factors.

One nuance from the Stage 2 results deserves a closer reading. Within the cultural block, the ESS attitudinal model (M5) outperforms the V-Dem institutional model (M1) on WAIC. The ESS predictors operationalise individual beliefs and norms — religiosity, gender attitudes, autonomy values — whereas the V-Dem predictors operationalise institutional

structures. Although both groups are conventionally classed as cultural in the demographic transition literature (Bongaarts and Watkins, 1996; Colleran, 2016), the small WAIC advantage of M5 over M1 suggests that what matters at the country level is the distribution of individual attitudes rather than the formal institutional architecture surrounding them. The evidence for this is modest — $\Delta\text{ELPD} = +0.4$ with $\text{SE} \approx 1.0$ — but it is consistent with the cultural-evolutionary account of fertility decline, in which the relevant unit of variation is the individual mind rather than the institution.

6.5 H4 — Total Causal Effect of Education on Completed Fertility

The H4 analysis turns from the country level to the individual level. The total effect of education on completed fertility is estimated under the partially linear regression model set out in Section 3.8 using DoubleML with ranger random forests (500 trees, minimum node size 5) as the nuisance learners. Five-fold cross-fitting with three repetitions was used; the reported estimate is the median across repetitions. The treatment is NVQ qualification level (0–4) measured at age 30, the outcome is total children at age 42, and the confounder vector comprises ten pre-treatment variables: father's social class at birth, mother's age, mother's age at first birth, parity, birth order, age-10 cognitive composite, age-10 social class, age-10 family income, age-16 social class, and the age-16 Malaise inventory score.

Figure 6.4 visualises the unadjusted education–fertility gradient by sex before any confounder adjustment. The pattern in this raw figure is striking: women's fertility declines steeply with educational qualifications, from 2.12 children at the lowest tier to 1.57 children at NVQ4+; men's fertility declines more modestly, from 1.73 children at the lowest tier to 1.51 at NVQ4+.

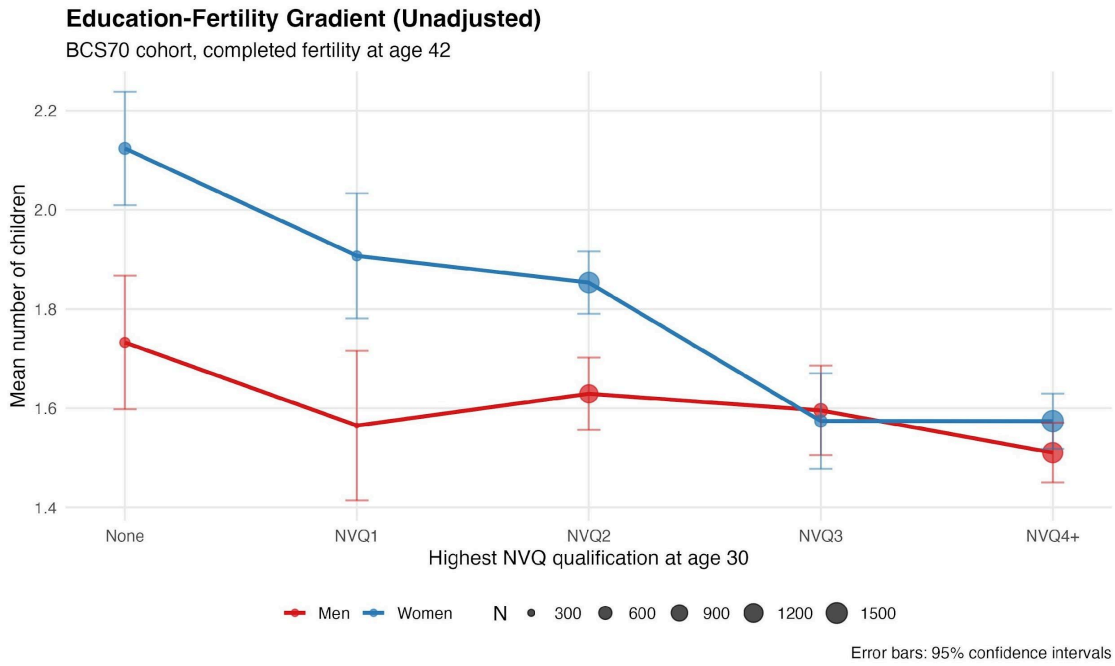


Figure 6.4. Unadjusted education–fertility gradient in the BCS70 cohort by sex. Mean completed fertility at age 42 by highest NVQ qualification at age 30. Point sizes scaled to cell size; error bars are 95% confidence intervals.

The DML total-effect estimates are reported in Table 6.4. Note that DML cross-fitting introduces stochastic variation between estimation runs; the figures reported here are taken from the headline total-effect specification. A separate run of the M0 mediation baseline produces an estimate of -0.140 for women (used in the decomposition arithmetic below), reflecting the random partition into folds. Substantively, the two estimates are indistinguishable, and the same conclusion holds under either.

Table 6.4. H4 — Total causal effect of education on completed fertility (DML partially linear regression). Estimates report the average causal effect of one additional NVQ level.

Sample	N	θ (SE)	95% CI	p-value
Women	1,461	-0.133 (0.028)	$[-0.188, -0.078]$	< 0.001
Men	1,081	-0.056 (0.033)	$[-0.120, +0.008]$	0.087
Pooled	2,542	-0.112 (0.022)	$[-0.154, -0.070]$	< 0.001

Note. θ is the average causal effect of a one-unit increase in NVQ level on the number of children at age 42. SEs are produced by the DML estimator with five-fold cross-fitting and three repetitions; the median across repetitions is reported. Confounder set: 10 pre-treatment variables for women and men models; 11 (including

sex) for the pooled model. The pooled model is reported for consistency with Chapter 3 and is not the principal inference vehicle.

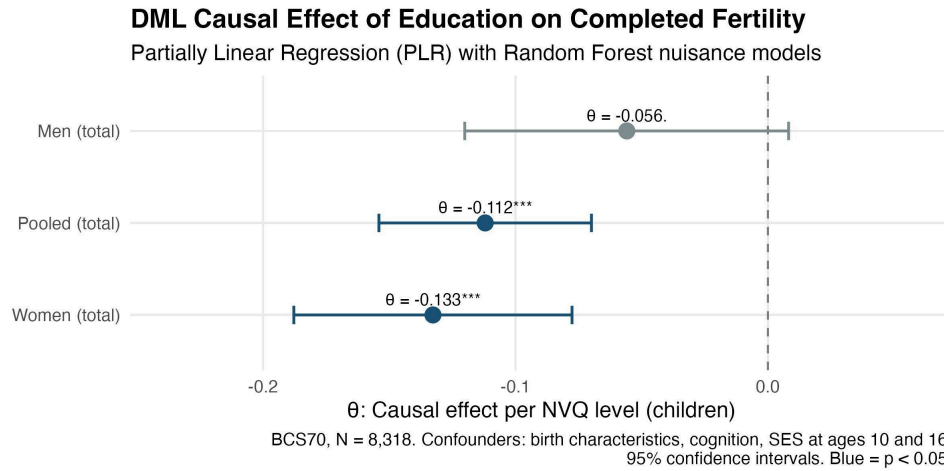


Figure 6.5. H4 — DML estimates of the total causal effect of education on completed fertility. Each point is the DML point estimate (PLR with random-forest nuisance learners); whiskers are 95% confidence intervals. Significant effects ($p < 0.05$) are shown in blue; the men's estimate (grey) is not statistically distinguishable from zero at conventional thresholds.

For women, an additional NVQ level reduces completed fertility by 0.133 children on average, with the 95% confidence interval running from -0.188 to -0.078 ($p < 0.001$). Across the full NVQ range (level 0 to level 4), this translates to approximately 0.53 fewer children for the most-educated women compared to women with no qualifications, conditional on the pre-treatment confounder set. For men, the estimated effect is smaller (-0.056) and not statistically distinguishable from zero at conventional thresholds ($p = 0.087$), with the 95% confidence interval crossing zero. The pooled estimate of -0.112 is statistically reliable but obscures the substantive sex difference.

Two points warrant emphasis. First, the women's effect survives extensive adjustment for high-dimensional pre-treatment confounding via the DML procedure, which controls for cognitive ability, family background, and adolescent functioning. The negative effect of education on fertility for British women born in 1970 is therefore not attributable to selection on the observed pre-treatment characteristics. Second, the much smaller and statistically

marginal effect for men is consistent with the broader sociological literature in which education's effect on fertility operates primarily through women — through delayed family formation, higher opportunity costs, and stronger career attachment — rather than through the male partner's education per se (Skirbekk, 2008). Given this sex difference, the subsequent mediation analysis is principally interpreted for women, with the men's results reported for completeness.

6.6 H4 — Sequential Mediation Decomposition

The mediation decomposition estimates the DML procedure four times for each sex, adding mediator groups sequentially to the confounder set as set out in Section 3.8. The total effect (M0) controls only for pre-treatment confounders. M1 adds the attitudinal mediators (*m_trad_gender*, *m_profamily*) measured at age 30; M2 adds the economic mediator (*log earnings* at age 42); and M3 adds the partnership mediator (*cohabiting partner* at age 42). The difference between successive estimates is interpreted as the indirect effect through the added mediator block. Sample sizes decline at each stage as more variables enter the complete-case requirement. Table 6.5 reports the sequential estimates.

Table 6.5. H4 — Sequential mediation decomposition for women and men. Each stage adds one mediator block to the DML conditioning set. The decline in N reflects complete-case attrition.

Stage	Mediators added	N	θ (SE)	p-value
M0 (women)	None (total effect)	1,461	−0.140 (0.028)	< 0.001
M1 (women)	+ Attitudinal	1,453	−0.135 (0.028)	< 0.001
M2 (women)	+ Earnings	1,046	−0.043 (0.031)	0.174
M3 (women)	+ Partnership	812	−0.075 (0.034)	0.028
M0 (men)	None (total effect)	1,081	−0.056 (0.033)	0.092

Stage	Mediators added	N	θ (SE)	p-value
M1 (men)	+ Attitudinal	1,075	−0.062 (0.033)	0.064
M2 (men)	+ Earnings	827	−0.091 (0.038)	0.018
M3 (men)	+ Partnership	678	−0.083 (0.042)	0.047

Note. Estimates from sequential DML with five-fold cross-fitting and three repetitions. The M0 estimate differs slightly from the headline total-effect estimate in Table 6.4 because of stochastic variation in cross-fitting partitions; substantive conclusions are identical under either.

For women, the sequential pattern is informative. The estimate barely moves from M0 to M1 (−0.140 to −0.135), indicating that adjusting for the attitudinal mediators absorbs only about 4% of the total effect. The largest drop is from M1 to M2 (−0.135 to −0.043, a reduction of 0.092 in absolute magnitude and a change in statistical significance from $p < 0.001$ to $p = 0.174$), indicating that the earnings mediator absorbs by far the largest share of the education effect. Adding the partnership mediator at M3 produces a partial reversal (back to −0.075 with $p = 0.028$), suggesting that partnership operates in the opposite direction to earnings — a finding I discuss in Section 6.7.

For men, the sequential pattern shows suppression rather than mediation. The estimate becomes more negative with the addition of each mediator block, from −0.056 at M0 to −0.083 at M3. This is the classical pattern of an explanatory factor that, once adjusted for, reveals a larger underlying effect — a feature of the data that I interpret carefully in the next section.

Figure 6.6 visualises the pathway decomposition for both sexes.

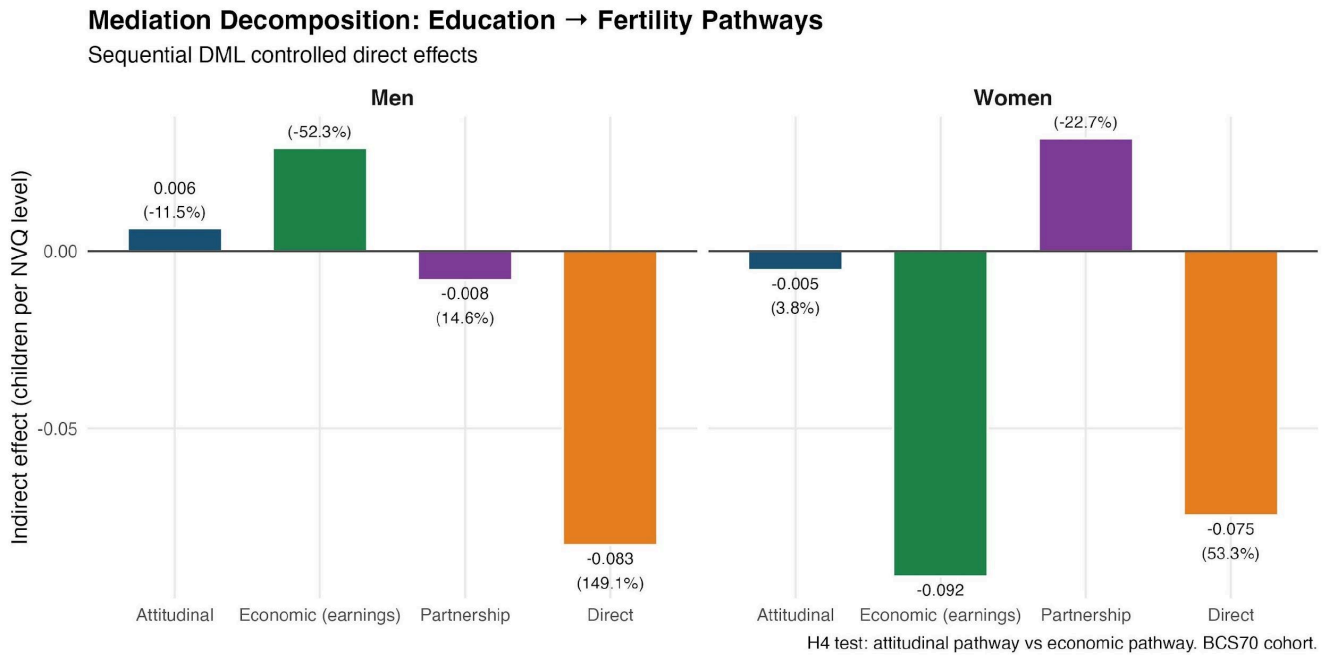


Figure 6.6. H4 — Mediation decomposition for women and men. Bars show the indirect effect through each pathway (attitudinal, economic, partnership) and the controlled direct effect remaining after sequential adjustment. Percentages are the pathway share of the total effect (M_0). For men, the direct effect exceeds the total effect (149%) because of suppression by the economic pathway.

6.7 H4 — Pathway-Specific Results and Test of H4

Table 6.6 reports the pathway-specific indirect effects and their share of the total effect for both sexes.

Table 6.6. H4 — Pathway decomposition by sex. Effects are the indirect effects through each pathway (computed by subtraction of sequential DML estimates); the direct effect is the residual after all mediator adjustment.

Pathway	Women (effect)	Women (% total)	Men (effect)	Men (% total)
Attitudinal	-0.005	3.8%	+0.006	-11.5%
Economic (earnings)	-0.092	65.6%	+0.029	-52.3%
Partnership	+0.032	-22.7%	-0.008	14.6%

Pathway	Women (effect)	Women (% total)	Men (effect)	Men (% total)
Direct (residual)	−0.075	53.3%	−0.083	149.1%

Note. For women, total effect = −0.140 (M0). For men, total effect = −0.056 (M0). Percentages are the pathway effect as a share of the total effect. Negative percentages reflect either suppression (men's economic and attitudinal pathways) or counter-acting indirect effects (women's partnership pathway). The pathway shares can sum to more than 100% when individual pathways operate in opposing directions, as is the case for both sexes.

The substantive conclusion for H4 is unambiguous and contrary to the hypothesis. H4 predicted that the effect of education on fertility would be mediated primarily by attitudinal pathways rather than economic ones. The data show the opposite. For women — the sample where the total effect is large and reliably estimated — the economic pathway (earnings) accounts for 65.6% of the total effect, the attitudinal pathway accounts for just 3.8%, and the direct effect represents the remaining 53.3%. The partnership pathway operates in the opposite direction (+0.032, recovering 22.7% of the total), consistent with the interpretation that highly educated women are more likely to be partnered at age 42 and that partnered status is positively associated with completed fertility, partially offsetting the negative earnings pathway.

For men, the pathway estimates are not directly interpretable as mediation under the standard assumptions, because the total effect is small and statistically marginal. The fact that the direct effect (−0.083) exceeds the total effect (−0.056) in absolute magnitude — the 149% figure in Table 6.6 — reflects suppression rather than mediation: the economic mediator masks rather than transmits the underlying effect of education on men's fertility. Substantively, this suggests that higher-educated men have higher fertility than their unadjusted estimates indicate, once their higher earnings are conditioned out. The data do not support a strong mediation story for men, and the H4 prediction (that attitudes dominate) is not borne out for either sex.

H4 is therefore not supported. The contemporary individual-level education–fertility relationship — at least within the British 1970 cohort, the most demanding test setting for H4 because of the BCS70's rich measurement of both attitudes and earnings — is mediated by economic rather than attitudinal pathways. The attitudinal pathway accounts for less than 4% of the effect for women and is positive-signed for men (operating against the negative total effect). This pattern is consistent with the quantity–quality framework of Becker (1960) and Kaplan (1996), in which extended education raises the opportunity cost of childbearing and shifts the optimal investment strategy toward fewer, more heavily invested children. It is inconsistent with the cultural-transmission account of Bongaarts and Watkins (1996) and Colleran (2016) under the strong individual-level reading in which transmitted attitudes are the principal proximate mechanism.

Two caveats temper this verdict. First, the attitudinal mediators (*m_trad_gender*, *m_profamily*) are measured at age 30, the same sweep as the treatment. Their proximity to the treatment means that the attitudinal pathway captures only that part of the education–attitude link that has consolidated by age 30, and may understate the true contribution of attitudes to fertility outcomes if attitudes continue to shape behaviour after age 30 in ways not fully captured by the age-30 measurement. This timing limitation, acknowledged in Section 3.8.2, is the principal threat to the H4 verdict; it cannot eliminate the conclusion that earnings dominate (the economic indirect effect of -0.092 is far larger than any plausible understatement of the attitudinal pathway) but it does warrant epistemic caution. Second, the BCS70 is a single British 1970 cohort. The H4 result generalises with confidence only to similar cohorts in similar institutional and labour-market settings. Both caveats are addressed substantively in Chapter 7.

6.8 Synthesis and Bridge to Chapter 7

Read together, the H3 and H4 findings present a coherent if uncomfortable picture for the cultural-transmission account of fertility decline. At the country level (H3), the cultural and economic blocks cannot be definitively separated at $N = 21$: the discovery–inference pipeline shows cultural indicators carrying slightly more in-sample importance (59.5% on SHAP) but the economic model winning the Bayesian comparison on WAIC, with all credible intervals crossing zero. The economic block holds the narrow inferential edge, but the small sample size precludes a definitive macro-level verdict in either direction.

At the individual level (H4), the verdict is sharper. The economic pathway (earnings) accounts for 65.6% of the total effect of education on women's completed fertility; the attitudinal pathway accounts for less than 4%. Education depresses women's fertility primarily through the opportunity cost of forgone earnings, not through the transmission of low-fertility attitudes. This is a falsification of H4 in its strongest form, and a substantive constraint on the cultural-transmission framework: whatever role transmitted attitudes may play in propagating low fertility across populations, they do not appear to be the principal individual-level mechanism through which education depresses fertility in the British 1970 cohort.

Taken together with the U-shape findings in Chapter 5, the H3 and H4 results help discriminate between the theoretical frameworks introduced in Chapter 2. The quantity–quality and opportunity-cost frameworks — both of which place economic mechanisms at the centre of the contemporary education–fertility relationship — receive the most support from this dissertation's evidence. The cultural-transmission framework receives mixed support at the country level and is contradicted at the individual level. The maladaptation framework is not directly tested by the analyses here, but the H4 finding that

the effect of education operates through realised opportunity costs rather than through evolved attitudinal mechanisms is more consistent with the rational-choice flavour of the quantity–quality account than with the psychological-mismatch flavour of the maladaptation account.

These theoretical implications, alongside the strengths and limitations of the empirical strategy and the directions for future research, are taken up in Chapter 7.